Homework 1

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Question 1: A First Look at Denoising Models

(a) Posterior. Both $p(x_0)$ and $p(x_1 \mid x_0)$ are Gaussian with mean linear in x_0, so (x_0, x_1) is jointly Gaussian and the posterior is Gaussian before any algebra; the work is to find its two parameters. By Bayes' rule, keeping only the factors that involve x_0,

$$p(x_0 \mid x_1) \;\propto\; p(x_0)\, p(x_1 \mid x_0) \;\propto\; \exp\left\{-\frac{x_0^2}{2\tau_0^2} - \frac{(x_1 - \sqrt{1-\beta}\, x_0)^2}{2\beta}\right\}.$$

Expanding the square and dropping $x_1^2/(2\beta)$, which is constant in x_0, the exponent becomes

$$-\frac{x_0^2}{2\tau_0^2} - \frac{(1-\beta)\, x_0^2 - 2\sqrt{1-\beta}\, x_0 x_1}{2\beta} \;=\; -\frac{1}{2}\underbrace{\left(\frac{1}{\tau_0^2} + \frac{1-\beta}{\beta}\right)}_{\lambda} x_0^2 \;+\; \underbrace{\frac{\sqrt{1-\beta}\, x_1}{\beta}}_{h}\, x_0.$$

Completing the square, any density proportional to $\exp\{-\frac{1}{2}\lambda x_0^2 + h x_0\}$ is $\mathcal{N}(h/\lambda,\, 1/\lambda)$. Here $\lambda = \big(\beta + (1-\beta)\tau_0^2\big)/(\beta\tau_0^2)$, so

$$\frac{1}{\lambda} = \frac{\beta\,\tau_0^2}{\beta + (1-\beta)\tau_0^2}, \qquad \frac{h}{\lambda} = \frac{\sqrt{1-\beta}\, x_1}{\beta} \cdot \frac{\beta\,\tau_0^2}{\beta + (1-\beta)\tau_0^2} = \frac{\sqrt{1-\beta}\,\tau_0^2\, x_1}{\beta + (1-\beta)\tau_0^2},$$

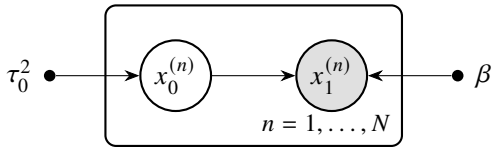
$$\boxed{x_0 \mid x_1 \;\sim\; \mathcal{N}\!\left(\frac{\sqrt{1-\beta}\,\tau_0^2}{\beta + (1-\beta)\tau_0^2}\, x_1,\;\; \frac{\beta\,\tau_0^2}{\beta + (1-\beta)\tau_0^2}\right).}$$

Check. Jointly $\mathbb{E}[x_1] = 0$, $\mathrm{Var}(x_1) = (1-\beta)\tau_0^2 + \beta$ by total variance, and $\mathrm{Cov}(x_0, x_1) = \mathbb{E}[x_0\, \mathbb{E}(x_1 \mid x_0)] = \sqrt{1-\beta}\,\tau_0^2$; Gaussian conditioning, mean $\mathrm{Cov}(x_0, x_1)\, x_1/\mathrm{Var}(x_1)$ and variance $\tau_0^2 - \mathrm{Cov}(x_0, x_1)^2/\mathrm{Var}(x_1)$, reproduces both boxed expressions.

The mean is $x_1/\sqrt{1-\beta}$, the inversion of the deterministic scaling, times $(1-\beta)\tau_0^2/\big((1-\beta)\tau_0^2 + \beta\big)$, the share of $\mathrm{Var}(x_1)$ due to signal rather than injected noise; the posterior therefore shrinks that inversion toward the prior mean 0. Its variance is free of x_1: an observation relocates the posterior without changing its spread.

(b) Limits. As $\beta \to 0$ the mean tends to x_1 and the variance to 0: the step is the identity, so x_1 pins down x_0 exactly and the posterior is a point mass. As $\beta \to 1$ the mean tends to 0 and the variance to τ_0^2: x_1 is pure noise, the likelihood is flat, and the posterior falls back on the prior $\mathcal{N}(0, \tau_0^2)$. Since $d\sigma^2/d\beta = \tau_0^4/\big(\beta + (1-\beta)\tau_0^2\big)^2 > 0$, the variance climbs monotonically between these limits, so β sets how much of x_0 survives the step.

(c) Graphical model. The statement describes one (x_0, x_1) pair; the plate shows the i.i.d. replication over N of them. Shaded $x_1^{(n)}$ is observed, $x_0^{(n)}$ is latent, and the fixed τ_0^2, β sit outside the plate, shared across all n.



Question 2: Discrete Conjugacy

(a) Posterior. Write $\alpha_0 = \sum_{v=1}^{V} \alpha_v$ and $B(\boldsymbol{a}) = \prod_{v=1}^{V} \Gamma(a_v) / \Gamma\big(\sum_{v=1}^{V} a_v\big)$, so the prior density on the simplex $\Delta_{V-1} = \{\boldsymbol{\pi} : \pi_v \geq 0, \sum_v \pi_v = 1\}$ is $p(\boldsymbol{\pi}) = B(\boldsymbol{\alpha})^{-1} \prod_{v=1}^{V} \pi_v^{\alpha_v - 1}$; equivalently $\int_{\Delta_{V-1}} \prod_{v=1}^{V} \pi_v^{a_v - 1}\, d\boldsymbol{\pi} = B(\boldsymbol{a})$ for any $\boldsymbol{a}$ with positive entries, an identity reused in part (b). The likelihood is where the counts appear. Writing one categorical draw as $p(y_i \mid \boldsymbol{\pi}) = \pi_{y_i} = \prod_{v=1}^{V} \pi_v^{\mathbf{1}\{y_i = v\}}$ and using conditional independence given $\boldsymbol{\pi}$,

$$p(y_1, \ldots, y_n \mid \boldsymbol{\pi}) = \prod_{i=1}^{n} \prod_{v=1}^{V} \pi_v^{\mathbf{1}\{y_i = v\}} = \prod_{v=1}^{V} \pi_v^{\sum_{i=1}^{n} \mathbf{1}\{y_i = v\}} = \prod_{v=1}^{V} \pi_v^{n_v},$$

so the data enter only through the counts $(n_1, \ldots, n_V)$, a sufficient statistic, and the likelihood is a product of powers of the π_v, matching the functional form of the prior. That is exactly why the family is closed under conditioning. Multiplying,

$$p(\boldsymbol{\pi} \mid y_1, \ldots, y_n) \;\propto\; \prod_{v=1}^{V} \pi_v^{\alpha_v - 1} \prod_{v=1}^{V} \pi_v^{n_v} \;=\; \prod_{v=1}^{V} \pi_v^{(\alpha_v + n_v) - 1}.$$

The right side is the Dirichlet kernel with parameters $\alpha_v + n_v > 0$; a density is pinned down by its kernel, so the normalizer is $B(\boldsymbol{\alpha} + \boldsymbol{n})^{-1}$ and

$$\boxed{\boldsymbol{\pi} \mid y_1, \ldots, y_n \;\sim\; \text{Dirichlet}(\alpha_1 + n_1,\; \ldots,\; \alpha_V + n_V).}$$

The prior parameters behave as pseudocounts: the data simply add n_v to α_v, and the concentration grows from α_0 to $\alpha_0 + n$.

(b) Posterior predictive. Let $\boldsymbol{\alpha}' = \boldsymbol{\alpha} + \boldsymbol{n}$, so $\alpha'_0 = \sum_u \alpha'_u = \alpha_0 + n$. Given $\boldsymbol{\pi}$ the next draw is independent of the previous ones, so conditioning on $\boldsymbol{\pi}$ and marginalizing it out,

$$p(y_{n+1} = v \mid y_{1:n}) = \int_{\Delta_{V-1}} p(y_{n+1} = v \mid \boldsymbol{\pi})\, p(\boldsymbol{\pi} \mid y_{1:n})\, d\boldsymbol{\pi} = \int_{\Delta_{V-1}} \pi_v\, p(\boldsymbol{\pi} \mid y_{1:n})\, d\boldsymbol{\pi} = \mathbb{E}[\pi_v \mid y_{1:n}],$$

the posterior mean of π_v. The extra factor of π_v raises the v-th exponent by one, replacing $\boldsymbol{\alpha}'$ with $\boldsymbol{\alpha}' + \boldsymbol{e}_v$, so that same identity applies:

$$\mathbb{E}[\pi_v \mid y_{1:n}] = \frac{1}{B(\boldsymbol{\alpha}')} \int_{\Delta_{V-1}} \prod_{u=1}^{V} \pi_u^{(\alpha'_u + \mathbf{1}\{u=v\}) - 1}\, d\boldsymbol{\pi} = \frac{B(\boldsymbol{\alpha}' + \boldsymbol{e}_v)}{B(\boldsymbol{\alpha}')} = \frac{\Gamma(\alpha'_v + 1)}{\Gamma(\alpha'_v)} \cdot \frac{\Gamma(\alpha'_0)}{\Gamma(\alpha'_0 + 1)} = \frac{\alpha'_v}{\alpha'_0},$$

where the $u \neq v$ factors cancel and $\Gamma(z+1) = z\,\Gamma(z)$ is used twice. Hence

$$\boxed{p(y_{n+1} = v \mid y_1, \ldots, y_n) = \frac{\alpha_v + n_v}{\alpha_0 + n}, \qquad \alpha_0 = \sum_{u=1}^{V} \alpha_u.}$$

These sum to one over v, since $\sum_v (\alpha_v + n_v) = \alpha_0 + n$. Rewriting

$$\frac{\alpha_v + n_v}{\alpha_0 + n} = \frac{n}{\alpha_0 + n} \cdot \frac{n_v}{n} \;+\; \frac{\alpha_0}{\alpha_0 + n} \cdot \frac{\alpha_v}{\alpha_0}$$

exhibits it as a convex combination of the maximum likelihood estimate n_v/n and the prior mean α_v/α_0, with α_0 an effective prior sample size: it tends to the MLE as $n \to \infty$ but keeps positive mass on every category even when $n_v = 0$. This is additive smoothing, which becomes Laplace smoothing when $\alpha_v = 1$.